Listing 10.9 The SsNode::directionOfMaxVariance method

```
function
SsNode::directionOfMaxVariance()
       #1
  maxVariance ←
0
#2
  directionIndex ←
0                                                          #3
  centroids ← this. getEntriesCentroids()
                          #4
  for i in {0..k-1} do
                 #5
    if varianceAlongDirection(centroids, i) > maxVariance
then                #6
      maxVariance ← varianceAlongDirection(centroids,
i)                      #7
      directionIndex ←
i                                                    #8
  return
directionIndex
      #9
```

#1 Method `directionOfMaxVariance` is defined on `SsNode`. It returns the index of the direction along which the children of a node have maximum variance.

#2/#3 Properly initialize the maximum variance and the index of the direction with max variance.

#4 Get the centroids of the items inside the node’s bounding envelope: for a leaf, those are the points held by the leaf, while for an internal node, the centroids of the node’s children.